Project Instructions for
Cryptographic Algorithms and AI-Based Security Application

Project Overview: Design and implement a security application that combines cryptographic algorithms, a user-friendly graphical interface, and at least one AI-based security feature.

1. Project Requirements

- Implement different cryptographic algorithms, including classical and modern encryption, hashing, and public-key cryptography. Examples include Caesar Cipher, Vigenère Cipher, AES, RSA, and SHA-256.
- Develop a Graphical User Interface (GUI) that allows users to select algorithms and perform applicable operations such as encryption, decryption, hashing, and verification.
- Integrate at least one AI-based security feature from the suggested options below. The feature must be implemented and demonstrated, not only described.
- Test the application with different scenarios, including invalid keys, modified messages, incorrect inputs, and successful operations. Evaluate correctness and performance.
- Prepare a report and presentation explaining the algorithms, AI feature, GUI design, testing results, and relevant security considerations.

2. Suggested AI-Based Features

Possible ideas include AI-based email spam detection, phishing detection, password strength analysis, anomaly detection, sensitive-information detection, suspicious-message classification, and AI-based cryptographic recommendations. These are only suggestions; teams are welcome to propose and implement other creative AI-based security features.

3. Team Size

Each project team must consist of 5 to 8 students. Any exceptional case or request to form a team outside this range must receive the instructor's approval in advance.

4. Assessment Rubric (10 Marks)

Assessment Criteria	Marks	What to Assess
Project Idea & AI Integration	2.0	Originality, usefulness, and working implementation of the AI feature.
Understanding of Cryptographic Algorithms	2.0	Correct understanding of algorithms, keys, encryption/decryption, and hashing.
Implementation & Testing	2.5	Correct functionality, successful test cases, error handling, and reliable results.
GUI Design & Usability	2.0	Clear, attractive, consistent interface and ease of use.
Presentation & Discussion	1.5	Clear explanation, live demonstration, and ability to answer questions.
TOTAL	**10.0**	

5. Important Notes

- The AI feature must work in the application; a placeholder button or a theoretical description is not sufficient.
- Students should understand the algorithms they use, even when relying on established software libraries.
- Testing should demonstrate both successful and unsuccessful cases, where applicable.
- The GUI should be functional and easy to navigate; visual design alone does not replace correct implementation.